

Zhenglong (Michael) Li

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SUMMARY

Applied Mathematics & Statistics student with a Computer Science minor, focused on computational and quantitative problem solving in computational geometry, scientific and high-performance computing, AI/ML, data analysis, and numerical methods. Background includes ongoing undergraduate research, the ISC 2026 Student Cluster Competition, work in geometry processing and time-series forecasting, and experience as a teaching assistant.

EDUCATION

2023 – present **Stony Brook University** (GPA: **3.85**/4.00)
B.S., Applied Mathematics & Statistics; Minor in Computer Science
Relevant coursework: Data Structures, Computational Geometry, Parallel Computing, Numerical Analysis, Probability & Statistics, Applied Linear Algebra, Applied Real Analysis, Complex Analysis, Differential Equations

RESEARCH & TECHNICAL PROJECTS

Long-Term Time-Series Forecasting Error Analysis (Summer 2026 – Present)

Conducted empirical analysis of direct-forecasting error propagation across multiple long-term time-series models and benchmark datasets. Reproduced and audited selected model runs against official configurations, standardized position-wise MSE/MAE error curves across multiple forecasting backbones, and built a unified analysis pipeline for comparing near-to-far error growth, cross-model rankings, and local/global alignment using correlation-based statistics. Organized experiment provenance, validation tables, figures, and research notes for group discussion and reporting.

ISC 2026 Student Cluster Competition (Nov 2025 – Jun 2026)

AI Training Optimization: Took a leading role in optimizing Qwen3-14B training with JAX/MaxText on multi-GPU HPC nodes, comparing batch size, precision, quantization, rematerialization, learning-rate scheduling, checkpoint behavior, and runtime logs. Helped document a final checkpoint-bearing run with 53.9% higher throughput than the starter reference (3283.6 Tokens/s/device) and final loss 0.629 under the 5-epoch task constraint.

HemeLB/HemePure Simulation: Took a leading role in HemeLB CPU-side work for Aneurysm-VIRTUAL blood-flow simulations, including Slurm job workflows, MPI rank-scaling experiments, monitoring/intrinsics comparisons, IPM profiling summaries, full-run result organization, and ParaView-based flow visualization videos.

On-site Competition and Presentation: Participated in the ISC 2026 Student Cluster Competition in Germany, contributing to final presentation preparation, technical Q&A support, and on-site team coordination.

Parallel Computing — SeaWulf HPC (Fall 2025)

Implemented three MPI-based matrix multiplication variants: 1D row-wise, Fox 2D block, and Strassen with a communication-aware layout. Added a ring token-passing mechanism to guarantee deterministic, rank-ordered output. Verified results against a serial implementation. Benchmarked strong scaling on SeaWulf for matrices up to 4096 x 4096 using up to 64 ranks; observed near-linear speedup for 1D and Fox and analyzed Strassen's recursion and communication trade-offs at higher core counts.

Polygon Guarding via Star-Shaped Decomposition (Spring 2026)

Designed and implemented a computational-geometry pipeline for decomposing simple polygons into star-shaped guardable regions. Combined ear-clipping triangulation, Hertel–Mehlhorn-inspired convex pre-merging, Lee–Preparata-inspired kernel updates, and half-plane clipping validation to produce kernel-certified guard regions across multiple polygon examples.

Topology-Aware Local Mesh Repair for AI-Generated 3D Models (Summer 2026 – Present)

Developed a topology-aware framework for repairing non-manifold edges in AI-generated 3D meshes while restricting geometric modifications to small defect neighborhoods. Classified local surface configurations using edge-incidence and connectivity structure, and designed a staged family of local reconstruction operators with fallback strategies for both canonical and irregular defect patterns.

Built an automated validation pipeline for synthetic fixtures and large batches of naturally generated meshes, checking manifoldness, boundary behavior, connectivity, and geometric validity before and after repair. Used systematic failure analysis to refine the supported defect classes and distinguish locally repairable structures from coupled or highly irregular configurations.

Data Analysis (Spring 2025)

Built a multiple regression pipeline for a simulated gene and environment dataset. Performed basic imputation, feature screening, and regularization-based variable selection (Lasso). Fit a square-root transformed response model, ran k-fold cross validation for model comparison, and used residual and QQ diagnostics to check assumptions. Delivered a reproducible R Markdown report within the figure and table limits.

Portfolio Optimization via Graphical Lasso (Spring 2026)

Led project coordination and implemented/cross-checked a quantitative portfolio-optimization pipeline using Graphical Lasso to estimate a sparse precision matrix for global minimum-variance portfolio construction. Built rolling-window backtests comparing optimized portfolios with equal-weight baselines, analyzed lambda selection and sparsity behavior over time, and evaluated out-of-sample performance using cumulative growth and Sharpe ratio.

Numerical Methods and Optimization Projects (Spring 2026)

Implemented numerical algorithms for optimization, approximation, integration, and differential equations, including simulated annealing for 3D path optimization, shooting and finite-difference methods for BVPs, Runge–Kutta methods for IVPs, polynomial interpolation/least-squares fitting, and Monte Carlo simulations.

EXPERIENCE

Teaching Assistant — SBU, AMS 345/CSE 355 Computational Geometry Fall 2026

Hold office hours and support students with computational geometry concepts and assignments; grade homework involving geometric algorithms and proofs; provide written feedback and coordinate grading decisions with the instructor and TA team.

Teaching Assistant — SBU, AMS 530 Principles of Parallel Computing Fall 2026

Serve as a teaching assistant for a graduate-level course in parallel computing; grade programming projects, reports, and experimental results; verify submitted code when needed on the university computing cluster; provide written feedback and coordinate grading decisions with the instructor and head TA.

Teaching Assistant — SBU, AMS 361 Applied Calculus IV Spring 2025

Held weekly office hours on ordinary differential equations, including first- and second-order equations and Laplace transforms; edited official homework solutions; graded assignments using consistent rubrics; coordinated regrades and FAQs with the professor and TA team.

Computational Geometry Research Group Jan 2026 – Present

Participate in research discussions and collaborative problem solving across computational geometry, including geometric graph problems, polygon structure, medial-axis-based reasoning, and geometric decom-

position. Recent work has explored local reference constructions for polygon fatness and related structural ambiguities, with emphasis on developing definitions, testing counterexamples, and refining algorithmic formulations through group discussion.

SKILLS

Programming	Python, Java, R, C, LaTeX, Bash/PowerShell
HPC & Tools	MPI/OpenMPI, Slurm, Linux, JAX/MaxText, Git, OpenAI Codex, Excel, MATLAB
Math & Visualization	Wolfram Mathematica, GeoGebra, Ipe, Mathcha, ParaView
Languages	Mandarin Chinese (native); English (professional)

HONORS & AWARDS

Robert J. and Kathryn B. Frey Scholarship (2025)	SBU Dean’s List (all semesters)
Global Excellence Scholarship	Stay on Track Scholarship
Second Prize, National Biology Competition (2022)	